

Praveen Kumar

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Location: Hyderabad

Professional Summary

Highly motivated Machine Vision and Embedded AI engineer with a strong passion for developing real-time inspection and monitoring systems, excited to contribute to Mavionix AI as a Junior AI Research Engineer or Embedded Systems Intern. Experienced in deploying computer vision models on edge devices and integrating thermal, ultrasonic, and optical sensors for defect detection and predictive maintenance systems, with a strong foundation in PyTorch, Deep Learning, and Computer Vision.

Technical Skills

- **Machine Learning:** PyTorch, Scikit-learn, TensorFlow, Keras
- **Computer Vision:** YOLOv8, OpenCV, Object Detection, Image Processing
- **Edge AI:** Raspberry Pi, Embedded Inference, Real-Time Processing
- **IoT & Embedded:** Arduino, Sensor Integration, Control Systems
- **Sensors:** LiDAR, Ultrasonic, Thermal, Optical, Gyroscope
- **Tools:** Git, Jupyter Notebook, Twilio API, CAD
- **Programming:** Python, Java, SQL

Education

Bachelor of Technology in Computer Science (AI & ML), Woxsen University, Hyderabad, 2022 – 2026

Key Projects

- **Railway AI Research (Patent Published):** Designed AI-enabled wheel inspection system using machine vision and sensor fusion, integrated LiDAR, thermal, ultrasonic, and optical sensors for defect detection, and developed microcontroller-based real-time diagnostic and IoT monitoring system.
- **Smart Crop Guard – Real-Time Edge AI Monitoring System:** Developed embedded machine vision system using YOLOv8 achieving sub-200 ms inference latency, built full AI pipeline including dataset preparation, training, and edge deployment, and integrated sensors, lights, and speakers with automated alerts via Twilio API.

- **AI-Powered Smart Irrigation System:** Developed EfficientNet-B0 model achieving 90%+ accuracy for crop-stage classification, built real-time inference pipeline on Raspberry Pi with camera and soil sensors, and designed offline irrigation decision logic using vision and sensor data.
- **Patented Dual-Axis Solar Tracking System:** Designed and patented dual-axis solar tracking system using LDR sensors for real-time orientation, developed motor control algorithms reducing alignment error by 25% and increasing energy capture by 40%, and built embedded control system integrating sensors and microcontroller-based tracking logic.
- **Smart Waste Segregation System:** Designed automated waste segregation system using Arduino-based control and sensor detection, developed embedded control logic for real-time classification and mechanical actuation, and built rotating separation mechanism for reliable material sorting.

Achievements & Publications

- **IEEE Publication** – Smart Crop Guard: AI-Enabled Wildlife and Adaptive Deterrence System
- **Patent Granted** – Sun Tracking Solar Panel System
- **Patent Published** – Wheel Condition Assessment Device (Application No:-202541067091)
- **Co-author** – IoT Applications Book Chapter Role of IoT In Communications (Taylor & Francis)
- **Treasurer** – IEEE Student Branch
- **Shortlisted** – MSME Zonal SIH 2024